SOA APIs

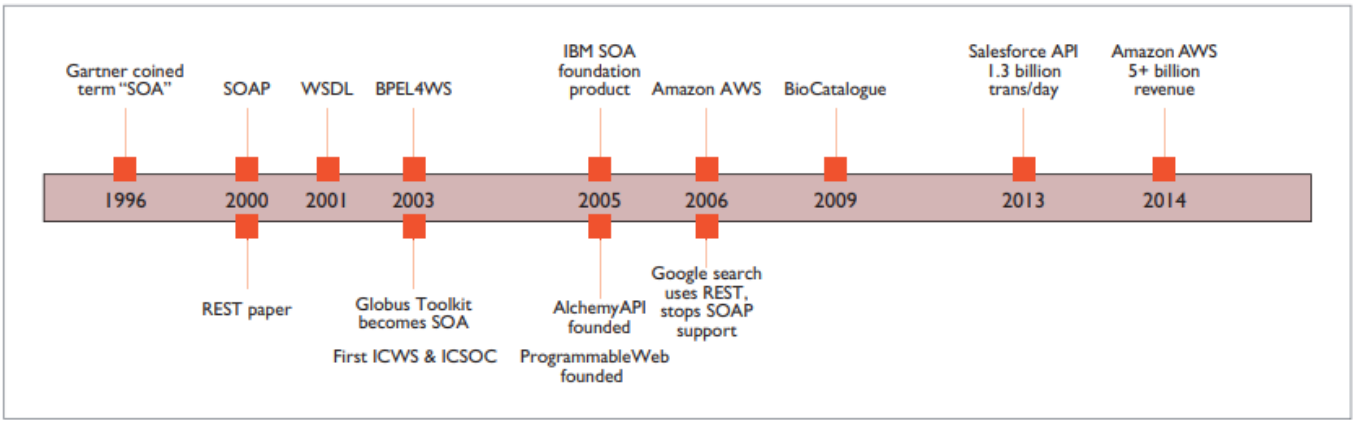

Figure 1. A brief history of services computing. After debuting in the early 2000s, services computing has transitioned from an enterprise interoperation technology to shape the Web API economy. (AWS = Amazon Web Services; BPEL4WS = Business Process Execution Language for Web Services; SOA = service-oriented architecture; WSDL = Web Services Description Language.)

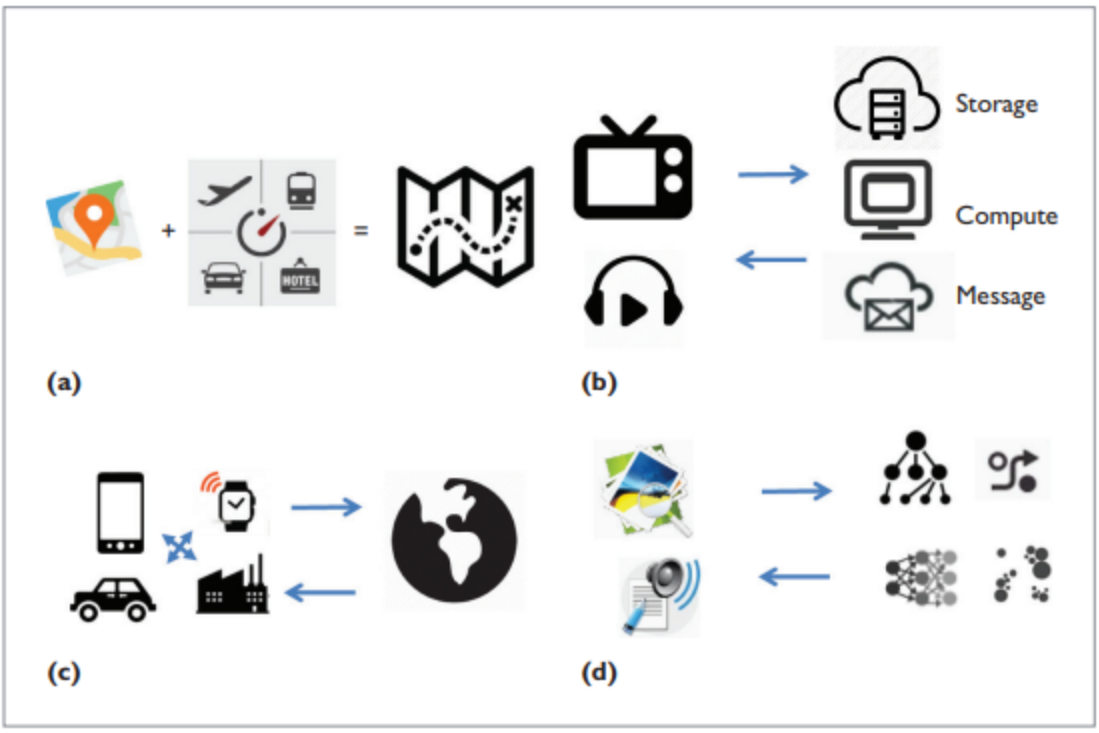

Figure 2. Web, cloud, mobile/Internet of Things (IoT), and machine learning applications are fully reliant on Web services/APIs. (a) Web (mashup); (b) cloud (infrastructure); (c) mobile and IoT; and (d) machine learning

Table 1. Topics of interest to the services computing community, and their relation to other computer science areas.

Topics	Related to computer science areas
Verification, composition	Software engineering
Semantics, security, and invocation framework	Web
Workflow, provenance, and data management	Databases
Distributed services, reliability, scalability, and performance	Systems
Automatic composition, recommendation	AI
Service interface, usability	Human-computer interface (HCI)
Optimization of service composition, quality	Operational research (OR)

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